Md Sameer

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.13
Sri Chaitanya Jr. Kalasala	Class XII (Telangana Board of Intermediate Education)	2024	97.50%
Sri Chaitanya High School	Class X (Telangana Board of Secondary Education)	2022	10.00%

PROJECTS

Extended xv6 Operating System Kernel (RISC-V):

- Implemented a **System-Call-Aware MLFQ process scheduler** with dynamic priority adjustment, starvation prevention, priority boosting, and runtime scheduling statistics.
- Built a virtual memory subsystem with lazy allocation, Clock page replacement for page eviction, frame management and disk-backed swapping.
- Simulated **RAID 0/1/5** for storage with FCFS/SSTF disk scheduling and parity-based fault tolerance, along with kernel instrumentation and user-space tests for system validation.

ISA Design and Evaluation for Specific Applications:

- Analyzed application bottlenecks and developed **custom RISC-V ISA instructions** using the **RVSS Simulator** to accelerate applications compared to baseline implementations.
- Implemented **CMOV (Conditional Move)** to reduce branch overhead and minimize **pipeline stalls and execution cycles** in conditional operations.
- Implemented **ADDC (Add with Carry)** to optimize **multi-precision arithmetic**, reducing instruction count and improving performance for large-scale computations.

Arbitrary Precision Calculator:

- Developed a **Java-based arbitrary precision calculator** supporting integer and floating-point arithmetic beyond native data type limits.
- Applied **object-oriented design principles** to improve modularity, reusability, and maintainability and Structured the project using **Git** for version control.

Query Optimization & Performance Analysis:

- Analyzed **SQL query performance** across varying query selectivities by testing **No Index**, **B-Tree and clustered indexes**, along with different index combinations, to identify effective indexing strategies.
- Evaluated **PostgreSQL query optimizer** decisions using **pg_hint_plan** by comparing execution times to determine **optimal plan-switch points** and execution strategies across different query workloads.

SKILLS

Programming Languages: C, C++, Python, SQL, Assembly (RISC-V)

Web / Development: HTML, CSS, JavaScript, PostgreSQL, Git, CMake

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Architecture, Theory of Computation, Software Development Fundamentals

Mathematics: Discrete Mathematics, Calculus, Probability and Random Variables, Introduction to Statistics, Linear Algebra

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA) | Dept of Computer Science

- Working as a teaching assistant for the course Introduction to Programming

Hospitality & Logistics Coordinator | Extra Mural Lectures (EML), IIT Hyderabad

- Coordinated hospitality and logistics for invited speakers during institute-level lecture events.
- Collaborated with organizing teams to ensure smooth event execution and effective speaker management.

Design Coordinator | Sunshine (Counselling Cell), IIT Hyderabad

- Designed newsletters, posters, and promotional material for awareness and student outreach.

Design Coordinator | SPIC MACAY, IIT Hyderabad

- Designed posters, brochures, and promotional content for cultural and heritage events.
- Worked with the organizing team to create visual material for student outreach.

EXTRACURRICULAR

Prayas RDC Volunteer (Rural Development Centre): Mentored government school students through educational initiatives, providing academic guidance, conducting learning sessions and encouraging personal growth through educational support programs.

NSS Volunteer: Participated in community service and social outreach initiatives across the campus.